

A lot is happening with self-replicating 3D printers too

The first affordable 3D printer was created by RepRap, which is capable of manufacturing parts required to duplicate itself, which can then be shared with anyone. This concept of self-replicating continues to cultivate new RepRaps.



Prusa i3 3D printer (Image courtesy: reprap.org)



iTopie RepRap printer (Image courtesy: reprap.org)



Delta 3 printer (Image courtesy: 3Dprint.com)



Production-ready BI V2.0 prototype (Image courtesy: kickstarter.com)

Of late, 3D printing in outer space has created a lot of buzz. A bunch of South African students, who call themselves Space Invaders, have recently developed a mobile, self-replicating 3D printer called Delta 3. It integrates delta geometry to create structures and parts with very high and accurate standards meant for space exploration.

Another self-replicating printer from Boots Industries, BI V2.0, was successfully funded on Kickstarter. This self-replicating, high-precision printer is a delta-style 3D printer featuring widely-available and self-replicating 3D-printed components.

Considering the latest in the array of RepRaps, the third generation of Prusa i3 was released, which offered new features on earlier generation models. iTopie, a derivative of Prusa i3, targets to achieve better timing for assembling RepRap 3D printer with strong focus on increasing rigidity, using a medium-density fibreboard (MDF) structure that is stable, durable and affordable.

Mishra. He says, "Companies, like Amazon, Microsoft and FedEx, are helping create a business for personalised items, such as phone covers, figurines, 3D-printed jewellery and many such products."

3D printers are fostering customisation as well, informs Vivekanand-havarma Datla, head-Manufacturing Engineering Solutions, Cyient Ltd. He says, "Personalised, custom-printed iPhone cases are just the beginning of an era of hyper-customisation, potentially achievable through 3D printing."

In robotics and DIY community.

Here too, 3D printing has been a boon. It is enabling printing of various small parts designed by makers for their projects. Such ease of fabrication would not be possible without 3D printing.

3D printing offers great amount of freedom to engineers

"From desktop to desk in a few minutes," summarises Shishodia the process of 3D printing. With 3D printing, an engineer can have a prototype of his design ready within a few min-

utes to a few hours by printing it on his desk.

3D printing aids in converting an idea into a physical product in the quickest possible time. Speed to market is the biggest gain for engineers using this technology, believes Datla. He notes, "There is a paradigm shift from 'design for manufacture' to 'manufacture anything that can be designed.'"

With 3D printing becoming faster and cheaper, engineers will have the luxury of modelling and prototyping more of their designs before production, feels Sidhant Pai, CEO Protoprint Solutions Pvt Ltd. He says, "The ability to physically test their models will likely lead to better and more usable products. Additionally, high-quality 3D printing will enable customised add-ons to user products, freeing product designers to factor in personalisation in their designs."

While there are desktop 3D printers available from popular companies, engineers also go for DIY kits that prove to be slightly cheaper and also give them an insight into the nuances of 3D printers during the course of the build, notes Shishodia. He says, "Open source 3D printers, like Prusa i3 and OrdBot Hadron, are quite popular among the DIY 3D-printer community."

Talking specifically about how specifications of today's printers could help engineers, Reddy says, "In FDM, techniques like auto-bed leveling help by reducing users' interventions. Algorithms that make the 3D printer's tool head move slowly at curves, intricate designs or small objects, increases the cool-down time and quality." He adds, "Different approaches of a machine's mechanical designs, such as coreXY mechanism, helical or delta-styled 3D printers, are opening up more possibilities for machine features."

Will 2015 be equally exciting for 3D printing

3D printing is a broad umbrella term and there have been a lot of